

Review for Midterm 1: Lectures 1-5

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Review for MT1

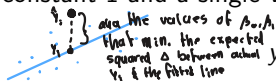
3 Q for midterm! 75 min

A4 Paper cheat!

- 1 Univariate Regression
- 2 Conditional Expectation Function, Population Regression
- 3 Regressions with Categorical Variables (dummies) → you should be able to construct a reg w/ just dummies & a how CEF
- 4 Oaxaca Decomposition (discrete/continuous X) → within/ata component
- 5 Frisch-Waugh Theorem and Omitted Variable Bias
- 6 Causal Model vs. Population Regression, IV-2SLS and Reduced Form - key assumptions
understand assumptions

Lecture 1 - Univariate Regression — be comfortable deriving the var

- Consider a population regression of outcome variable y_i on a constant 1 and a single variable x_i :



$$y_i = \beta_0 + \beta_1 x_i + U$$

$$y_i = \beta_0 + \beta_1 x_i + U$$

$$\Rightarrow y_i - \beta_0 - \beta_1 x_i = U_i$$

$$\Rightarrow U_i^2 = (y_i - \beta_0 - \beta_1 x_i)^2$$

$$\Rightarrow E[U_i^2] = E[(y_i - \beta_0 - \beta_1 x_i)^2] \quad (1)$$

$$\text{in which } (\beta_0, \beta_1) = \underset{b_0, b_1}{\operatorname{argmin}} E[(y_i - b_0 - b_1 x_i)^2]$$

$\rightarrow$ the values of β_0, β_1 that minimize the squared residual: b_0, b_1

- By solving the first-order conditions, we have:

$$\beta_0 = E[y_i] - \beta_1 E[x_i]$$

$$\beta_1 = \frac{\operatorname{cov}(y_i, x_i)}{\operatorname{var}(x_i)}$$

(2)

- Note the FOCs are equivalent to:

$$U_i \perp \beta_0, \beta_1 \text{ \& } y_i?$$

$$E[U_i] = 0, E[U_i x_i] = 0 \quad \star$$

(3)

which you have shown in question 1 of PS2.

- Make sure you can prove (2), and understand the role of the constant. Without a constant, the fitted regression may not cross the point $(E[X], E[Y])$ – “fit the mean”. See question 1 of PS3.

(2) When given pop. reg: $y_i = \beta_0 + \beta_1 x_i + u$ s.t. $(\beta_0, \beta_1) = \arg\min_{b_0, b_1} E[(y_i - b_0 - b_1 x_i)^2]$

to find the FOC:

$$M(b_0, b_1) = E[(y_i - b_0 - b_1 x_i)^2]$$

$$= E[y_i^2 - 2b_0 y_i - 2b_1 x_i y_i + b_0^2 + 2b_0 b_1 x_i + b_1^2 x_i^2]$$

$$= E[y_i^2] - 2b_0 E[y_i] - 2b_1 E[x_i y_i] + b_0^2 + 2b_0 b_1 E[x_i] + b_1^2 E[x_i^2]$$

$$\Rightarrow \begin{cases} \frac{\partial M(b_0, b_1)}{\partial b_0} = -2E[y_i] + 2b_0 + 2b_1 E[x_i] \end{cases}$$

$$\Rightarrow \begin{cases} \frac{\partial M(b_0, b_1)}{\partial b_1} = -2E[x_i y_i] + 2b_0 E[x_i] + 2b_1 E[x_i^2] \end{cases}$$

$$(y_i - b_0 - b_1 x_i)(y_i - b_0 - b_1 x_i)$$

$$\Rightarrow y_i^2 - \underline{y_i b_0} - \underline{y_i b_1 x_i}$$

$$- \underline{y_i b_0} + b_0^2 + b_0 b_1 x_i$$

$$- \underline{b_1 x_i y_i} + b_1 b_0 + b_1^2 x_i^2$$

$$\Rightarrow y_i^2 - 2y_i b_0 - 2y_i x_i b_1$$

Now to solve we set them both equal to 0 (remember: $\arg\min_{b_0, b_1} E[(y_i - b_0 - b_1 x_i)^2]$)

$$\Rightarrow \begin{cases} -2E[y_i] + 2b_0 + b_1 E[x_i] = 0 & \textcircled{1} \\ -2E[x_i y_i] + 2b_0 E[x_i] + 2b_1 E[x_i^2] = 0 & \textcircled{2} \end{cases} \Rightarrow \begin{cases} -E[y_i] + b_0 + b_1 E[x_i] = 0 & \textcircled{1} \\ -E[x_i y_i] + b_0 E[x_i] + b_1 E[x_i^2] = 0 & \textcircled{2} \end{cases}$$

$$\Rightarrow b_0 = E[y_i] - b_1 E[x_i] \text{ from } \textcircled{1} \Rightarrow \text{plug into } \textcircled{2}: -E[x_i y_i] + (E[y_i] - b_1 E[x_i])E[x_i] + b_1 E[x_i^2] = 0$$

$$\Rightarrow b_1 (E[x_i^2] - E[x_i]^2) = E[x_i y_i] - E[x_i] E[y_i] \Rightarrow b_1 = \frac{E[x_i y_i] - E[x_i] E[y_i]}{E[x_i^2] - E[x_i]^2}$$

$$\Rightarrow b_1 = \frac{\text{Cov}(y_i, x_i)}{\text{Var}(x_i)} \text{ and } b_0 = E[y_i] - b_1 E[x_i]$$

(3) Why are $E[u_i] = 0$ & $E[u_i x_i] = 0$ when for $y_i = \beta_0 + \beta_1 x_i + u_i$
s.t. $m(\beta_0, \beta_1) = \arg \min_{b_0, b_1} E[(y_i - b_0 - b_1 x_i)^2]$?

Lecture 1 - Univariate Regression

$$\bar{y}_i = \frac{1}{n} \sum_{i=1}^n y_i$$

- Given a random sample $\{y_i, x_i\}_{i=1}^n$, we estimate an OLS regression of y_i on 1 and x_i :

$$(\hat{\beta}_0, \hat{\beta}_1) = \underset{b}{\operatorname{argmin}} \frac{1}{n} \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2 \quad (4)$$

$$\rightarrow \hat{\beta}_0 = \bar{y}_n - \hat{\beta}_1 \bar{x}_n$$

$$\hat{\beta}_1 = \frac{\sum_i (y_i - \bar{y}) x_i}{\sum_i (x_i - \bar{x}) x_i} = \frac{\hat{\operatorname{cov}}(x_i, y_i)}{\hat{\operatorname{var}}(x_i)}$$

which uses the fact that

$\hat{\operatorname{cov}}(x_i, y_i) = \sum_i (y_i - \bar{y})(x_i - \bar{x}) = \sum_i (y_i - \bar{y}) x_i = \sum_i (x_i - \bar{x}) y_i$. It

$\uparrow$ just a constant in sample and drop it b/c $\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x}) \cdot c$

Lecture 1 - Univariate Regression

- Be able to show consistency
- Give 95% CI but barely asymptotic

is equivalent to write:

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x}) y_i}{\sum_i (x_i - \bar{x}) x_i}$$

plug in $y_i = \beta_0 + \beta_1 x_i + u_i$:

$$\begin{aligned}\hat{\beta}_1 &= \frac{\sum_i (x_i - \bar{x}) (\beta_0 + \beta_1 x_i + u_i)}{\sum_i (x_i - \bar{x}) x_i} \\ &= \beta_0 * 0 + \beta_1 * 1 + \frac{\sum_i (x_i - \bar{x}) u_i}{\sum_i (x_i - \bar{x}) x_i} \\ &= \beta_1 + \underbrace{\frac{\sum_i (x_i - \bar{x}) u_i}{\sum_i (x_i - \bar{x}) x_i}}\end{aligned}\tag{5}$$

Lecture 1 - Univariate Regression

- Is $\hat{\beta}_1$ a consistent estimate for β_1 in the population regression (1)?
Focus on (5), and show consistency via the Law of Large Numbers:

$$\begin{aligned}\frac{1}{n-1} \sum_i x_i(u_i - \bar{u}) &\rightarrow^p \text{cov}(x_i, u_i) = 0 && \text{from FOC of pop. reg } u_i \perp x_i \\ &&& \text{b/c } E[x_i u_i] = 0 \\ \frac{1}{n-1} \sum_i (x_i - \bar{x})x_i &\xrightarrow{\text{plim}} \text{var}(x_i) \\ \hat{\beta}_1 = \beta_1 + \frac{\frac{1}{n-1} \sum_i (x_i - \bar{x})u_i}{\frac{1}{n-1} \sum_i (x_i - \bar{x})x_i} &\xrightarrow{\text{converges by LLN}} \beta_1 + \frac{\text{cov}(x_i, u_i)}{\text{var}(x_i)} = \beta_1 \\ \text{and } \hat{\beta}_0 &\rightarrow^p E[Y] - \beta_1 E[X] = \beta_0\end{aligned}$$

note $\text{cov}(x_i, u_i) = E[x_i u_i] - E[x_i]E[u_i] = 0$ because of the orthogonality condition ?(3) in the population regression

Lecture 1 - Univariate Regression

- Construct a 95% confidence interval for β_1 , using the asymptotic distribution:

$$\sqrt{n}(\hat{\beta}_1 - \beta_1) = \frac{\frac{1}{\sqrt{n}} \sum_i (x_i - \bar{x}) u_i}{\hat{\text{var}}(x_i)} \rightarrow^d N(0, \frac{\sigma^2}{\text{var}(x_i)}) \quad (6)$$

which uses the Central Limit Theorem and assumes homoscedasticity:

$$\begin{aligned} \frac{1}{\sqrt{n}} \sum_i (x_i - \bar{x}) u_i &\rightarrow^d N(0, \text{var}((x_i - E[x_i]) u_i)) \\ \text{var}((x_i - E[x_i]) u_i) &= E[(x_i - E[x_i])^2 u_i^2] - \underbrace{E[(x_i - E[x_i]) u_i]^2}_{=0} \\ &= E_X[(x_i - E[x_i])^2 \underbrace{E[u_i^2 | X]}_{=\sigma^2}] \quad \text{under homoscedasticity} \\ &= \text{var}(x_i) \times \sigma^2. \end{aligned}$$

Lecture 1 - Univariate Regression

95% confidence interval (assume sample size $n > 30$ and t-distribution is approximately normal):

replace pop. st. dev. w/
asymptotic

$$\left(\hat{\beta}_1 - 1.96 * SE(\hat{\beta}_1), \hat{\beta}_1 + 1.96 * SE(\hat{\beta}_1) \right)$$

$$SE(\hat{\beta}_1) = \frac{1}{\sqrt{n}} \times \sqrt{\frac{\hat{\sigma}^2}{\text{var}(x_i)}} \approx \frac{1}{\sqrt{n}} \times \sqrt{\frac{\sum_i \hat{u}_i^2}{\sum_i (x_i - \bar{x})^2}}$$

might be $\frac{1}{n-2}$ here
(D.F. = 2) but since
 $n \rightarrow \infty$ its using.

Lecture 2 - CEF

- CEF $E[Y|X]$ is a function of X that represents the expectation of Y conditional on random variables X .

$$E[Y|X] = \int_y y f_{Y|X}(y) dy$$

- Let $\epsilon = Y - E[Y|X]$. Use the law of iterated expectations to show:

$$E[\epsilon] = 0, E[\epsilon|X] = 0, \text{ and } E[\epsilon h(X)] = 0 \text{ for any function of } X.$$

- CEF itself doesn't have to be linear. But we often use a linear approximation to CEF, which is our population regression. The "infeasible" (or population) regression estimator solves:

$$\min_{\beta} E[(y_i - x_i' \beta)^2] \quad (7)$$

$$\frac{\partial}{\partial \beta} = -2E[x_i(y_i - x_i' \beta)] = 0 \text{ at } \beta^* \quad \text{only def. w/ vector to know}$$

$$\Rightarrow \beta^* = E[x_i x_i']^{-1} E[x_i y_i]$$

$\epsilon \perp X$
are
mean independent

in lecture 2 we showed $x_i' \beta^*$ is the best **linear** approximation to CEF $E[y_i | x_i]$. We sometimes write the population regression as $E^*[y_i | x_i] = x_i' \beta^*$, where β^* solves (7).

- The feasible OLS estimator minimizes the sum of squared residuals (SSR) in the sample: $E^*[y|x] = x'\beta^*$

$$\hat{\beta} = \underset{\beta}{\operatorname{argmin}} \sum_{i=1}^N (y_i - x_i' \beta)^2 \quad (8)$$

$$\begin{aligned} &= \left[\sum_{i=1}^N x_i x_i' \right]^{-1} \sum_{i=1}^N x_i y_i \\ &= \beta^* + \left[\sum_{i=1}^N x_i x_i' \right]^{-1} \sum_{i=1}^N x_i u_i \end{aligned} \quad (9)$$

from (9) you could show **consistency** and asymptotic distribution of $\sqrt{n}(\hat{\beta} - \beta^*)$. You should be comfortable reaching (9), which represents the relationship between OLS estimator in sample and the population regression estimator.

Lectures 2-3: Regressions with Dummy Variables

- Suppose you are given a categorical variable x_i that can take a finite number of values $\{1 \dots K\}$. The conditional expectation function would be linear:

$$E[y_i | x_i] = \sum_{k=1}^K E[y_i | x_i = k] \times D_{ik} \quad (10)$$

Handwritten notes:
- $x = D \in \{0, 1\}$
- $E[y|x] = \begin{cases} E[y|x=1] \\ E[y|x=0] \end{cases}$
- indicator (arrow pointing to D_{ik})

where $D_{ik} := 1[x_i = k]$

Handwritten notes:
- ex. f/ PS2
- of example
- where dummy = 1 = 1 year?

(10) is equivalent to a population regression of y_i on dummies $\{D_{ik} : k = 1 \dots K\}$ for all possible values of x_i :

Handwritten notes:
- You'll define a
- series of dummies
- for each possible
- interaction

$$y_i = \sum_{k=1}^K \beta_k \times D_{ik} + \epsilon_i$$

Handwritten notes:
- interaction term (under $\beta_k \times D_{ik}$)

Handwritten note: PS2!

in which $\begin{bmatrix} \beta_1 \\ \dots \\ \beta_K \end{bmatrix} = \underset{b}{\operatorname{argmin}} E[(y_i - \sum_k b_k D_{ik})^2] \quad (11)$

Lectures 2-3: Regressions with Dummy Variables

Let's prove each $\beta_k = E[y_i | D_{ik} = 1] = E[y_i | x_i = k]$ without using vector notations: the first order condition is

$$\frac{\partial}{\partial b_k} = -2E[(y_i - \sum_{k'} b_{k'} D_{ik'}) D_{ik}] = 0$$

satisfied at $b = \beta$

$$\text{note } (y_i - \sum_{k'=1}^K \beta_{k'} D_{ik'}) D_{ik} = \begin{cases} 0 & \text{if } D_{ik} = 0 \\ y_i - \beta_k & \text{if } D_{ik} = 1 \end{cases}$$

b/c mutually exclusive dummies

So the FOC for β_k becomes:

$$E[(y_i - \sum_{k'} b_{k'} D_{ik'}) D_{ik}] = E[y_i - \beta_k | D_{ik} = 1] * Pr(D_{ik} = 1) = 0$$

implicitly using law of iterated expectations

$$\Rightarrow E[y_i - \beta_k | D_{ik} = 1] = 0 \quad (12)$$

$$\Rightarrow \beta_k = E[y_i | D_{ik} = 1] = E[y_i | x_i = k] \quad (\text{PS2})!$$

↳ explains what β_k means if pop. reg!
very important

Lectures 2-3: Regressions with Dummy Variables

- Note although the regression does not include a constant, we have also fit the mean: *it comes to early*

$$\begin{aligned} E[y_i] &= E_x[E[y_i|x_i]] && \text{apply LIE} \\ &= \sum_{k=1}^K E[y_i|x_i = k] \times Pr(x_i = k) \\ &= \sum_k \beta_k \times E[D_{ik}] && \text{be comfy w/ this!} \\ & && \text{conditional mean} \end{aligned} \tag{13}$$

- If you include a constant, you need to drop one of the dummy to ensure the covariates are not colinear with each other. (Note: need $E[x_i x_i']$ to be an invertible/nonsingular K-by-K matrix, the columns

Lectures 2-3: Regressions with Dummy Variables

should be linearly independent from each other). Suppose we drop the category $x_i = 1$:

$$y_i = \alpha + \sum_{k=2}^K \beta_k \times D_{ik} + \epsilon_i$$

in which
$$\begin{bmatrix} \alpha \\ \beta_2 \\ \dots \beta_K \end{bmatrix} = \underset{a,b}{\operatorname{argmin}} E[(y_i - a - \sum_{k=2}^K b_k D_{ik})^2] \quad (14)$$

Again use the FOCs of the problem (14) to show:

cell w/ non-col

$$\alpha = E[y_i | x_i = 1] \quad \text{mean earnings for non-col}$$

$$\beta_k = E[y_i | x_i = k] - E[y_i | x_i = 1]$$

↳ diff. mean earnings relative to non-col group

Review question #3 in PS2.

Lecture 3: Oaxaca Decomposition - Discrete X

- Now consider what happens with 2 groups, a and b . Suppose we are interested in the difference in outcome y between these two groups, and we observe characteristics $x_i = [D_{1i}, D_{2i}, \dots, D_{Gi}]'$ indicate G mutually exclusive categories (e.g., education).
- We ask: how much of the difference $\bar{y}^a - \bar{y}^b$ can be explained differences in composition (x) between these groups? For example, group a may have higher earnings on average because there are more college-educated workers in a than b . How much can the earning gap be explained by differences in education?
- Based on what we just reviewed about regressions on dummies (see equation (13)), consider two separate regressions:

$$\text{Group a: } y_i = \sum_{g=1}^G \hat{\beta}_g^a \times D_{ig} + e_i$$

mean earning f/cat. g
in group a

$$\text{Group b: } y_i = \sum_{g=1}^G \hat{\beta}_g^b \times D_{ig} + u_i$$

... within cat. g in group b

Lecture 3: Oaxaca Decomposition - Discrete X

From (13) we know the regression with dummies for every possible value of x can fit the mean. The sample analogue of that is:

$$\bar{y}^a = (\bar{x}^a)' \hat{\beta}^a$$

$$\bar{y}^b = (\bar{x}^b)' \hat{\beta}^b$$

in which $\bar{x}^a = \begin{bmatrix} \frac{1}{N^a} \sum_{i \in a} D_{i1} \\ \dots \\ \frac{1}{N^a} \sum_{i \in a} D_{iG} \end{bmatrix} = \begin{bmatrix} \bar{p}_1^a \\ \dots \\ \bar{p}_G^a \end{bmatrix}$, and $\hat{\beta}^a = \begin{bmatrix} \bar{y}_1^a \\ \dots \\ \bar{y}_G^a \end{bmatrix}$
(shown via FOC as we did for the population regression). So we have:

$$\bar{y}^a = \sum_g \bar{p}_g^a \bar{y}_g^a, \text{ and } \bar{y}^b = \sum_g \bar{p}_g^b \bar{y}_g^b \quad (15)$$

$$\bar{y}^b = (\bar{x}^b)' \hat{\beta}^b$$

- **Oaxaca decomposition** of $\bar{y}^b - \bar{y}^a$: how much does the gap in earnings between $b =$ public sector and $a =$ private sector employees can be explained by differences in education levels?

Lecture 3: Oaxaca Decomposition - Discrete X

- We can answer this by constructing a “counterfactual world” where group b (the gov’t workers) had the same distribution across categories as group a (the private sector workers) but within each category, the mean for group b remains unchanged. Under this counterfactual, the mean for group b would be:

$$\bar{y}_{counterf}^b = \sum_g \bar{p}_g^a \times \bar{y}_g^b = (\bar{x}^a)' \hat{\beta}^b \quad (16)$$

reweight people in b by the prop.

This “reweighted counterfactual” allows us to decompose the wage gap as follows:

$$\begin{aligned} \bar{y}^b - \bar{y}^a &= (\bar{y}^b - \bar{y}_{counterf}^b) + (\bar{y}_{counterf}^b - \bar{y}^a) \\ &= \underbrace{(\bar{x}^b - \bar{x}^a)' \hat{\beta}_b}_{\text{between}} + \underbrace{(\bar{x}^a)' (\hat{\beta}^b - \hat{\beta}^a)}_{\text{within}} \end{aligned}$$

- Review Question 4 in PS2 or the example in lecture 3. Be able to define a counterfactual, and point out which part represents the “between” component.

Lecture 3b: Oaxaca decomposition - continuous X

DFL not in midterm 1 (but will come back)

- We can also apply the decomposition technique to continuous covariates. Lecture 3b provides an example that explains the pay gap between natives and immigrants by education (as a continuous variable).
- Let x_i denote a vector of covariates (including a constant, continuous/discrete covariates). Estimate an OLS regression of y_i on x_i separately for each group a, b :

$$\hat{\beta}^a = \operatorname{argmin}_{\beta} \sum_{i \in a} (y_i - x_i' \beta)^2 \quad (17)$$

$$\hat{\beta}^b = \operatorname{argmin}_{\beta} \sum_{i \in b} (y_i - x_i' \beta)^2$$

Recall with a constant in $x_i = (1, x_{2i}, \dots, x_{Ki})'$, the linear regression can fit the mean.

$$\bar{y}^a = (\bar{x}^a)' \hat{\beta}^a \quad (18)$$

$$\bar{y}^b = (\bar{x}^b)' \hat{\beta}^b$$

Lecture 3b: Oaxaca decomposition - continuous X

- Oaxaca-Blinder Decomposition:

$$\begin{aligned}\bar{y}^b - \bar{y}^a &= (\bar{x}^b)' \hat{\beta}^b - (\bar{x}^a)' \hat{\beta}^a \rightarrow (\bar{x}^b)' \hat{\beta}^b - (\bar{x}^a)' \hat{\beta}^a + (\bar{x}^a)' \hat{\beta}^a \\ &= \underbrace{(\bar{x}^b - \bar{x}^a)' \hat{\beta}^b}_{\text{between}} + \underbrace{(\bar{x}^a)' (\hat{\beta}^b - \hat{\beta}^a)}_{\text{within}} \\ &= \underbrace{(\bar{x}^b)' (\hat{\beta}^b - \hat{\beta}^a)}_{\text{within}} + \underbrace{(\bar{x}^b - \bar{x}^a)' \hat{\beta}^a}_{\text{between}}\end{aligned}$$

- See the estimates for separate regressions in Lecture 3b for example.

Lecture 3b: Oaxaca decomposition - continuous X

- Lecture 3b also considers a pooled regression that is estimated on the observations from group a and group b. Define $D_i = 1[i \in b]$ and fit an OLS in sample:

$$\hat{y}_i = \hat{\beta}_1 + \hat{\beta}_2 x_i + \hat{\gamma}_1 D_i \quad (19)$$

with the inclusion of group dummy D_i , this regression fits the mean of the outcome for both group a and group b : (Hint: FOC w.r.t. β_1 and γ_1)

$$\begin{aligned} \bar{y}^a &= \hat{\beta}_1 + \bar{x}^a \hat{\beta}_2, \quad \bar{y}^b = \hat{\beta}_1 + \bar{x}^b \hat{\beta}_2 + \hat{\gamma}_1 \\ \Rightarrow \bar{y}^b - \bar{y}^a &= (\bar{x}^b - \bar{x}^a) \hat{\beta}_2 + \hat{\gamma}_1 \end{aligned}$$

Lecture 3b: Oaxaca decomposition - continuous X

not in 3a but makes
sense after learning and
applying Frisch-Waugh

- What does the pooled estimate $\hat{\beta}_2$ converge to? Apply Frisch Waugh (to the population regression):

$$\beta_2 = \frac{\text{cov}(y_i, \tilde{x}_i)}{\text{var}(\tilde{x}_i)}$$

$$\tilde{x}_i = x_i - E^*[x_i | 1, D_i]$$

$$= x_i - (\mu^a + (\mu^b - \mu^a)D_i)$$

from reg of
dummies

mean outcome of baseline group a

$$\text{note } E[\tilde{x}_i] = 0$$

$$\text{thus } \text{cov}(y_i, \tilde{x}_i) = E[y_i \tilde{x}_i] = E_D[E[y_i \tilde{x}_i | D]]$$

$$= p * E[y_i \tilde{x}_i | b] + (1 - p) * E[y_i \tilde{x}_i | a].$$

num in reg. cov. β_1/β_2^a

You can also apply the law of iterated expectations on the denominator, and show β_2 as a weighted avg of β_2^a, β_2^b in the separate regressions.

practice showing this weighted avg
to apply Frisch-Waugh & decompose

Lecture 4: Frisch-Waugh Theorem

- We posit a population regression:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + u_i \quad (20)$$

$$\text{where } \beta = \underset{b}{\operatorname{argmin}} E[(y_i - x_i' b)^2]$$

- What is the population coefficient β_2 ? By Frisch-Waugh Theorem

$$\beta_2 = \frac{\operatorname{cov}(y_i, \xi_i)}{\operatorname{var}(\xi_i)}$$

$$\xi_i = x_{2i} - E^*[x_{2i} | 1, x_{1i}, x_{3i}]$$

$\xi_i = x_{ki} - E^*[x_{ki} | x_{(-k)i}]$ is the residual over the projection of x_{ki} on *all* other covariates. Interpret β_k = coefficient from univariate regression of y_i on x_{ki} , after “partialling out” other x 's.

- We have shown this in class. The key element of the proof is the orthogonality condition in the auxiliary regression:

$$\begin{aligned} x_{2i} &= \gamma_0 + \gamma_1 x_{1i} + \gamma_3 x_{3i} + \xi_i \\ E[\xi_i] &= E[\xi_i x_{1i}] = E[\xi_i x_{3i}] = 0 \\ \Rightarrow \operatorname{cov}(y_i, \xi_i) &= \beta_2 \operatorname{var}(\xi_i) \end{aligned}$$

b/c x_1, x_2, x_3
uncorrel. w/ ξ_i

Lecture 4: Frisch-Waugh Theorem

- PS3 asks you to prove FW in sample. Hint: use the orthogonality condition (FOC's) in the OLS.

Lecture 4: Omitted Variable Bias

- What happens if you forget a regressor?

$$y_i = \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + u_i$$

Suppose we don't include x_{3i} ?

- auxiliary regression: $x_{3i} = \pi_1 x_{1i} + \pi_2 x_{2i} + \xi_i$, where $E[\xi_i x_{1i}] = E[\xi_i x_{2i}] = 0$
- Plug the auxiliary regression into the population regression:

$$\begin{aligned} y_i &= \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 (\pi_1 x_{1i} + \pi_2 x_{2i} + \xi_i) + u_i \\ &= (\beta_1 + \beta_3 \pi_1) x_{1i} + (\beta_2 + \beta_3 \pi_2) x_{2i} + (\beta_3 \xi_i + u_i) \end{aligned}$$

importantly, the new residual $\beta_3 \xi_i + u_i$ is orthogonal to (x_{1i}, x_{2i}) .
What does that imply?

- $(\beta_1^o, \beta_2^o) := (\beta_1 + \beta_3 \pi_1, \beta_2 + \beta_3 \pi_2)$ satisfy FOC for population regression of y_i on (x_{1i}, x_{2i}) ! OVB

$\cdot \text{if } \beta_3 = 0$
 $\cdot \pi_2 = 0$

$\Rightarrow$ no OVB
 $\hookrightarrow$ you can tell if $\pi_2 = 0$ by looking at aux reg., apply Frisch-Waugh to verify $\text{cov}(\pi_2, x_2) = 0$

Lecture 4: Omitted Variable Bias

- Conclusion: If $y_i = \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + u_i$ and we don't include x_{3i} , the coefficient on x_{2i} is:

$$\beta_2^o = \beta_2 + \beta_3 \pi_2$$

where π_2 is the coefficient on x_{2i} from the regression of the omitted variable x_{3i} on the remaining x 's:

$$x_{3i} = \pi_1 x_{1i} + \pi_2 x_{2i} + \xi_i$$

- Going one step further, what is π_2 ? There is no omitted variable bias if $\pi_2 = 0$. Apply Frisch-Waugh again on the auxiliary regression, we have

$$\pi_2 = \frac{\text{cov}(x_{3i}, x_{2i} - E^*[x_{2i}|x_{1i}])}{\text{var}(x_{2i} - E^*[x_{2i}|x_{1i}])}$$

where we residualize x_2 over the other regressor x_{1i} . What if $x_{1i} = 1$? Then

$$\begin{aligned} E^*[x_{2i}|1] &= E[x_{2i}] \\ \pi_2 &= \frac{\text{cov}(x_{3i}, x_{2i})}{\text{var}(x_{2i})} \end{aligned}$$

Lecture 5: Population Regression vs. Causal Model

- Population Regression Model (PRM):

$$y_i = \beta_0 + \beta_1 x_{1i} + u_i = x_i' \beta + u_i$$

$$\text{in which } \beta = \operatorname{argmin}_b E[(y_i - x_i' b)^2] \quad (21)$$

- PRM: β is defined by f.o.c.: $E[x_i(y_i - x_i' \beta)] = 0$. That is, the residual $u_i := y_i - x_i' \beta$ satisfies $E[x_i u_i] = 0$. We often write PRM as $E^*[y_i | x_i] = x_i' \beta$.
- Causal model: Imagine we are thinking about a manipulation of x_{1i} . We ask “what is the **causal effect** of a change in x_{1i} , holding constant all other factors?” To address this, we assume for simplicity

$$y_i = \beta_0 + \beta_1 x_{1i} + u_i \quad (22)$$

i.e., the causal effect of x_1 is linear, with additive effect of all other factors. We can assume $E[u_i] = 0$, but **we cannot be sure** that $u_i \perp x_{1i}$. We may well have:

$$E[x_{1i} u_i] \neq 0 \quad (23)$$

Lec 5
very important
philosophically

Lecture 5: Population Regression vs. Causal Model

From OVT, population regression coefficient is

$$\begin{aligned}\beta_1^{PRM} &= \frac{E[(x_{1i} - E[x_{1i}]) y_i]}{E[(x_{1i} - E[x_{1i}])^2]} \\ &= \beta_1 + \lambda \\ \lambda &= E[(x_{1i} - E[x_{1i}])^2]^{-1} E[(x_{1i} - E[x_{1i}]) u_i]\end{aligned}\tag{24}$$

replace w/ model of interest (here w/ causal model)

where $\lambda \neq 0$ if $E[x_{1i} u_i] \neq 0$. You may interpret it as omitted variable bias: $u_i = \alpha_i + \nu_i$, $E[x_i \nu_i] = 0$ but $E[x_i \alpha_i] \neq 0$, α_i is omitted from the regression.

Lecture 5: Instrument Variable

- Since we just covered lecture 5, I won't repeat the details here. Here is what you need to review.
- Assumptions you need for a valid instrument: $E[z_i x_i] \neq 0$, $E[z_i u_i] = 0$.
- 2SLS cookbook. Show the 2nd stage that replaces x_i by $\text{pred}x_i$ works! You can get a consistent estimate for the causal effect b/c of the exclusion restriction.
- Show reduced form:

$$\beta_1 = \frac{\delta_1}{\pi_1}$$

by plugging in $x_i = \pi_0 + \pi_1 z_i + \xi_i$ to the causal model.

- As always, be comfortable with the sample analogue.

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